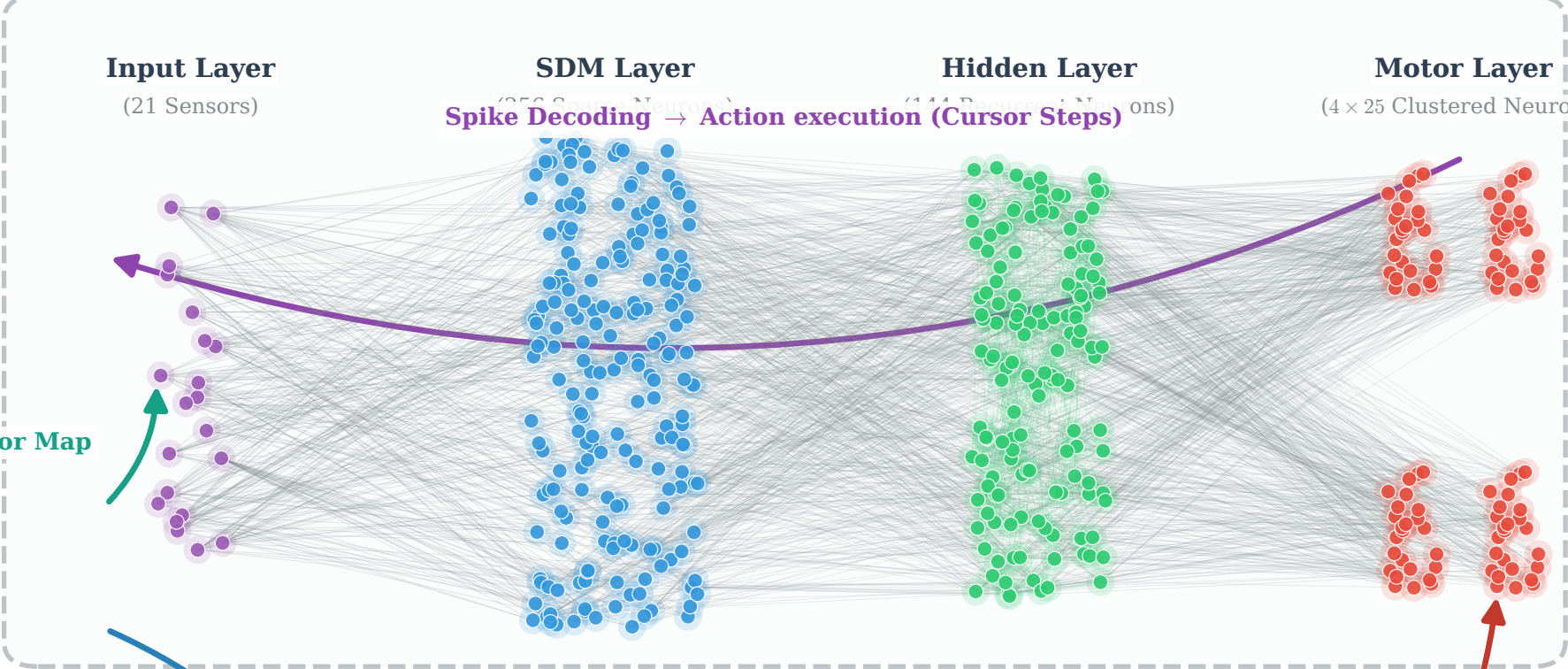


The Organoid (*in silico* Metabolic-Izhikevich Spiking Neural Network)



Exteroceptive Sensor Map

State  $s_t$   
Reward  $r_t$

Global Activity Regulator (Homeostatic Reset) \& Inhibitory Homeostasis \& Structural Plasticity

Motor Quadrant  
Injection  $I_{stim}$

Automated Experimentalist (Outer Agent)

Dueling Double DQN + Prioritized Exp. Replay

$$Q^*(s_t, a_t) = V(s_t) + \left( A(s_t, a_t) - \frac{1}{|\mathcal{A}|} \sum A(s_t, a) \right)$$